

Decision Tree

- Decision Trees do not require feature scaling.
- The first node is called the **Root Node**
- Each internal node contains a **condition** (True/False) used to split the data.
- The final nodes are called **Leaf Nodes**, and they contain the prediction (class or value).

Example of a root node:

```
petal_length < 2.45  
gini = 0  
samples = 150  
values = [50, 50, 50]  
class = setosa
```

Where:

- **samples:** Number of training samples reaching that node.
- **values:** Number of samples from each class.
- **class:** The majority class predicted at that node.

The tree follows the condition:

- If the condition is **True**, move to the left child.
- If the condition is **False**, move to the right child.
- Continue until reaching a **leaf node**, which gives the final prediction.

Gini Index (Gini Impurity)

The **Gini Index** measures how pure or impure a node is.

- **Pure node:** All samples belong to one class.
 - Gini = 0
- The larger the Gini value, the more mixed the classes are.

- The algorithm always tries to make Gini as small as possible after each split.

In other words:

- Lower Gini → Higher purity.
- Higher Gini → More impurity.

Model Interpretation

White-Box Models

These models are easy to interpret.

- We can understand **why** the model made a particular decision.
- Example: **Decision Tree**.

Black-Box Models

These models are difficult to interpret.

- We know the prediction but not exactly how the model reached it.
- They often achieve higher accuracy but have lower interpretability.

Examples:

- Neural Networks
- Random Forest

How a Decision Tree Chooses a Split

For every feature:

1. Try different possible split points.
2. Compute the Gini impurity after each split.
3. Choose the split that gives the **minimum Gini impurity**.

The goal is to produce child nodes that are as pure as possible.

Stopping Condition

The tree keeps splitting until a stopping condition is met.

Common stopping conditions include:

1. **Maximum Depth (max_depth)**

- Limits how deep the tree can grow.

2. **Maximum Features (max_features)**

- Limits the number of features considered when searching for the best split.

3. **Maximum Leaf Nodes (max_leaf_nodes)**

- Limits the total number of leaf nodes.

4. **Minimum Samples per Leaf (min_samples_leaf)**

- Every leaf must contain at least this many samples.

Regularization

Without restrictions, the tree may continue splitting until nearly every leaf is pure (Gini = 0).

This usually leads to **overfitting**.

To prevent overfitting, we use regularization by limiting the tree using parameters such as:

- max_depth
- max_features
- max_leaf_nodes
- min_samples_leaf

These constraints help the model generalize better to unseen data.

Prediction Process

1. Start at the **root node**.
2. Evaluate the condition.
3. Move left or right depending on whether the condition is True or False.
4. Repeat until reaching a **leaf node**.
5. The leaf node provides the final prediction.

Classification Evaluation Metrics

Accuracy

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Predictions}}$$

Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

- Measures **how many of the predicted positives are actually positive**.
 - Out of all the samples the model predicted as **True/Positive**, how many were actually true?

Recall (Sensitivity)

$$\text{Recall} = \frac{TP}{TP + FN}$$

- Measures **how many of the actual positive cases the model correctly identified**.
 - Out of all the actual positive samples, how many did the model detect?

Note:

- High Recall → Low FN (False Negatives).
- Important when missing a positive case is costly (e.g., disease detection).

F1-Score

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- The **harmonic mean** of Precision and Recall.
- Gives **equal weight** to Precision and Recall.

Confusion Matrix

TP (True Positive)

- The prediction is **correct**, and the model predicted **Positive**.

FP (False Positive)

- The prediction is **incorrect**.
- The model predicted **Positive**, but the actual class is **Negative**.

FN (False Negative)

- The prediction is **incorrect**.
- The model predicted **Negative**, but the actual class is **Positive**.

TN (True Negative)

- The prediction is **correct**.
- The model predicted **Negative**, and the actual class is **Negative**.